GLYCOGEN STORAGE DISEASES

- **Autosomal recessive**

- **12 types, all resulting in abnormal glycogen metabolism and an accumulation of glycogen within cells. Periodic acid–Schiff stain identifies glycogen and is useful in identifying these diseases.**

VON GIERKE DISEASE (TYPE I)

- Glucose-6-phosphatase deficiency
- **Findings:** increased Glycogen in liver, increased blood lactate, increased triglycerides and uric acid (Gout), and hepatomegaly. Glycogen structure normal
- **Treatment:** frequent oral glucose/cornstarch
- ; avoidance of fructose and galactose Impaired gluconeogenesis and glycogenolysis

- Deficiency of hepatic glucose -6-phosphatase produces a profound hypoglycemia,lactic acidosis and hepatomegaly
- Glycogen deposits found in the liver. Glucose-6-phosphatase stimulates glycogen synthesis and glycogenolysis is inhibited.

POMPE DISEASE (TYPE II)

- Lysosomal acid α-1,4-glucosidase with α-1,6-glucosidase activity (acid maltase)
- **Findings:** Cardiomegaly, hypertrophic cardiomyopathy, hypotonia, exercise intolerance, and systemic findings lead to early death.

- This disease is different from others because the deficient enzyme resides in the lysosomes and is responsible for digesting glycogen like material accumulating in endosomes. Thus indigestible substances accumulate as inclusion bodies . The tissues having glycogen stores are mostly affectes . With infantile onset massive cardiomegaly is usually the cause of death which occurs before 2 years of age.

CORI DISEASE (TYPE III)

- Debranching enzyme (α-1,6-glucosidase) deficiency
- **Findings:** Milder form of von Gierke , mild hypoglycemia , liver enlargement . Short outer branches single glucose residue at outer branch

MCARDLE DISEASE (TYPE V)

- Skeletal muscle glycogen phosphorylase (Myophosphorylase) deficiency
- **Findings**: increased glycogen in muscle, but

muscle cannot break it down leading to painful Muscle cramps, and Myoglobinuria (red urine) associated with with strenuous exercise, and arrhythmia from electrolyte abnormalities.

- Blood glucose levels typically
- unaffected
- McArdle = Muscle

- Exercise intolerance during initial phase of high intensity exercise
- Muscle cramping
- Due to myophosphorylase def patient unable to breakdown glycogen to glucose -6-p in muscles .